

Network Intrusion Detection System (IDS) Using Snort

CodeAlpha Cyber Security Internship – Task 4

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Abstract

This project demonstrates the implementation of a Network Intrusion Detection System using Snort on Kali Linux. A custom ICMP detection rule was created and tested successfully.

Objectives

Install Snort, configure custom rules, monitor traffic, generate ICMP packets, and verify alert generation.

Tools Used

Kali Linux, Snort 3.12.2.0, Nano Editor, Terminal, Ping Utility.

Implementation

The following screenshots document the complete implementation process.

```
kali@kali: ~  
Session Actions Edit View Help  
Setting up liblognorm5:amd64 (2.1.0-1)...  
Setting up libdaq3 (3.0.24-0kali1)...  
Setting up snort-rules-default (3.12.2.0-0kali1)...  
Setting up snort-common-libraries (3.12.2.0-0kali1)...  
Setting up rsyslog (8.2604.0-4)...  
Created symlink '/etc/systemd/system/syslog.service' → '/usr/lib/systemd/system/rsyslog.service'.  
Created symlink '/etc/systemd/system/multi-user.target.wants/rsyslog.service' → '/usr/lib/systemd/system/rsyslog.service'.  
Setting up liblua5.1-2:amd64 (2.1.0+openresty20251030-1+b2)...  
Setting up snort (3.12.2.0-0kali1)...  
snort.service is a disabled or a static unit, not starting it.  
Processing triggers for man-db (2.13.1-1)...  
Processing triggers for kali-menu (2026.2.2)...  
Processing triggers for libc-bin (2.42-13)...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
  
(kali@kali)-[~]  
$ snort -V  
  
--* Snort++ <*-  
o'')~  
''''~  
Version 3.12.2.0  
By Martin Roesch & The Snort Team  
http://snort.org/contact#team  
Copyright (C) 2014-2026 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using DAQ version 3.0.24  
Using libpcap version 1.10.6 (64-bit time_t, with TPACKET_V3)  
Using LuaJIT version 2.1.1761786044  
Using LZMA version 5.8.3  
Using OpenSSL 3.6.1 27 Jan 2026  
Using PCRE2 version 10.46 2025-08-27  
Using ZLIB version 1.3.1  
  
(kali@kali)-[~]  
$
```

Figure 1: Snort Installation and Version Verification

```
kali@kali: ~  
Session Actions Edit View Help  
Using LuaJIT version 2.1.1761786044  
Using LZMA version 5.8.3  
Using OpenSSL 3.6.1 27 Jan 2026  
Using PCRE2 version 10.46 2025-08-27  
Using ZLIB version 1.3.1  
  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.163.3/24 brd 192.168.163.255 scope global dynamic noprefixroute eth0  
        valid_lft 473sec preferred_lft 473sec  
    inet6 fe80::ba53:6795:bfe5:7e2e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5b:51:cd brd ff:ff:ff:ff:ff:ff  
    inet 192.168.0.107/24 brd 192.168.0.255 scope global dynamic noprefixroute eth1  
        valid_lft 82374sec preferred_lft 82374sec  
    inet6 fe80::ca1c:7e0e:a14f:a70d/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
    link/ether f6:77:e1:c1:86:81 brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever  
5: br-c2f6b63b97fe: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default  
    link/ether 22:a9:05:f4:28:2a brd ff:ff:ff:ff:ff:ff  
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-c2f6b63b97fe  
        valid_lft forever preferred_lft forever  
    inet6 fe80::20a9:5ff:fef4:282a/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
6: veth901c269aif2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-c2f6b63b97fe state UP group  
    default  
    link/ether 4e:13:91:71:38:91 brd ff:ff:ff:ff:ff:ff link-netnsid 0  
    inet6 fe80::4c13:91ff:fe71:3891/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
7: veth7687aeaif2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-c2f6b63b97fe state UP group  
    default  
    link/ether 4a:64:7c:f2:f0:22 brd ff:ff:ff:ff:ff:ff link-netnsid 1  
    inet6 fe80::4864:7cff:fef2:f022/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
$
```

Figure 2: Network Interface Identification (eth0)

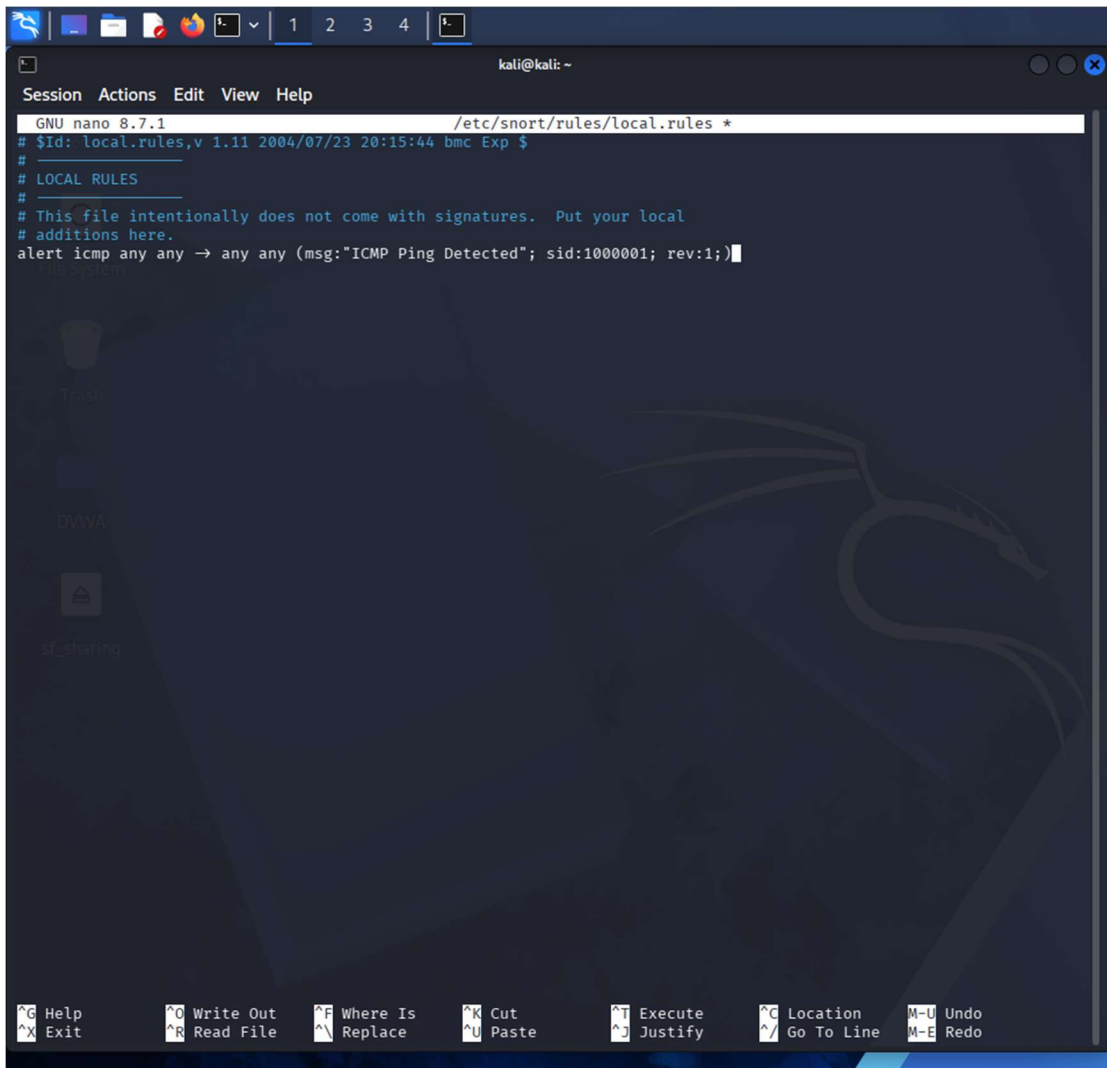


Figure 3: Custom ICMP Detection Rule

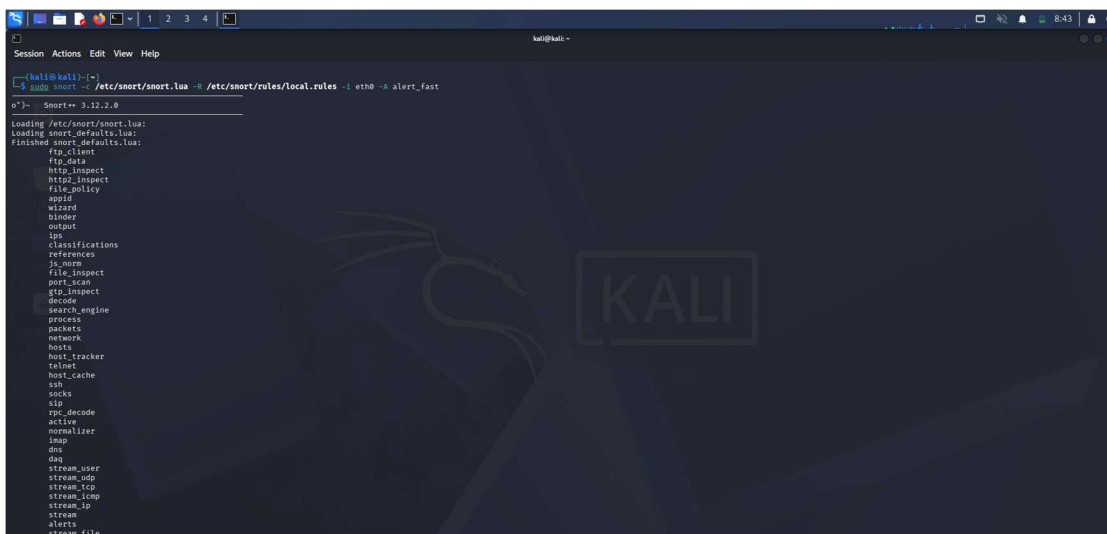


Figure 4: Snort Monitoring Started

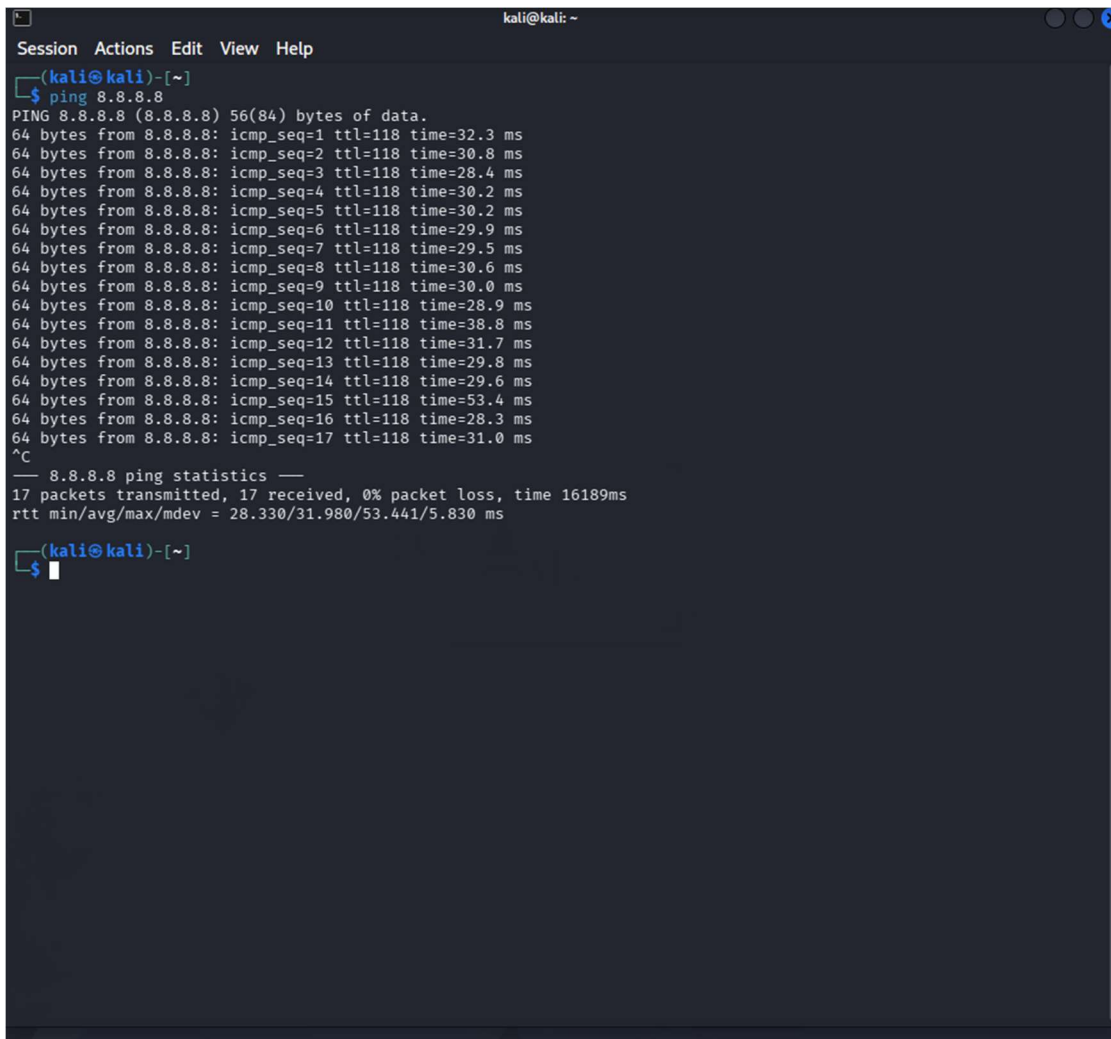
A terminal window titled 'kali@kali: ~' with a menu bar containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali@kali)-[~]'. The user enters '\$ ping 8.8.8.8'. The output shows 17 successful ping responses from 8.8.8.8, each with 64 bytes of data, TTL=118, and various response times. After pressing Ctrl-C (^C), the terminal displays '8.8.8.8 ping statistics' and summary statistics: '17 packets transmitted, 17 received, 0% packet loss, time 16189ms' and 'rtt min/avg/max/mdev = 28.330/31.980/53.441/5.830 ms'. The prompt returns to '\$'.

Figure 5: ICMP Traffic Generation using Ping

```
Session Actions Edit View Help
06/21-08:49:46.458918 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:46.489151 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:46.491546 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:47.490420 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:48.468633 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:48.497520 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:49.478421 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:49.508007 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:50.472227 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:50.499511 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:51.478532 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:51.503582 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:52.475135 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:52.518085 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:53.479663 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:53.508677 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:54.479674 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:54.509648 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:55.526873 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:55.558100 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:56.527922 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:57.531941 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:57.568090 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:58.518720 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:58.564855 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:59.541941 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:59.574737 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:50:00.564160 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:50:00.565556 *** [1:1000000:1] "ICMP Ping Detected" *** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
-- [0] eth1

Packet Statistics
daq
received: 55
analyzed: 55
allow: 55
rx_bytes: 6996

codec
total: 55 (100.000%)
arp: 10 (< 18.182%)
eth: 55 (100.000%)
icmp: 30 (< 54.545%)
ipnet: 45 (81.818%)
udp: 15 (< 27.273%)

Module Statistics
ac_full
```

Figure 6: Snort Alert Detection Output

Results

Snort successfully detected ICMP traffic generated using the ping command. The custom rule triggered alerts confirming correct IDS operation.

Conclusion

The Network IDS was successfully implemented and tested. Snort detected ICMP packets and generated alerts according to the configured rule.